

C PROGRAMMING LAB

82 Programs — Complete Code Collection

Programs are arranged in the corrected continuous numbering from 1 to 82.

Note: Each program is standalone. Compile and run one program at a time. Programs using `math.h` may require `-lm` when compiling with GCC.

A. Basic C Programs

1. Add two numbers

```
#include <stdio.h>

int main() {
    int a, b;
    scanf("%d %d", &a, &b);
    printf("Sum = %d", a + b);
    return 0;
}
```

2. Difference between two integers

```
#include <stdio.h>

int main() {
    int a, b;
    scanf("%d %d", &a, &b);
    printf("Difference = %d", a - b);
    return 0;
}
```

3. Multiply and divide two integers

```
#include <stdio.h>

int main() {
    int a, b;
    scanf("%d %d", &a, &b);
    printf("Multiplication = %d\n", a * b);

    if (b != 0)
        printf("Division = %.2f", (float)a / b);
    else
        printf("Cannot divide by zero");

    return 0;
}
```

4. Addition, subtraction, multiplication, quotient and remainder

```
#include <stdio.h>

int main() {
    int a, b;
    scanf("%d %d", &a, &b);

    printf("Addition = %d\n", a + b);
    printf("Subtraction = %d\n", a - b);
    printf("Multiplication = %d\n", a * b);

    if (b != 0) {
        printf("Quotient = %d\n", a / b);
        printf("Remainder = %d\n", a % b);
    } else {
        printf("Division by zero is not possible\n");
    }

    return 0;
}
```

5. Celsius to Fahrenheit

```
#include <stdio.h>

int main() {
    float celsius, fahrenheit;
    scanf("%f", &celsius);
    fahrenheit = (celsius * 9 / 5) + 32;
    printf("Fahrenheit = %.2f", fahrenheit);
    return 0;
}
```

6. Absolute value of a long integer

```
#include <stdio.h>
#include <stdlib.h>
```

```

int main() {
    long int n;
    scanf("%ld", &n);
    printf("Absolute value = %ld", labs(n));
    return 0;
}

```

7. Learn to use math.h header functions

```

#include <stdio.h>
#include <math.h>

int main() {
    double n;
    scanf("%lf", &n);

    printf("Square root = %.2f\n", sqrt(n));
    printf("Power = %.2f\n", pow(n, 2));
    printf("Absolute value = %.2f\n", fabs(n));

    return 0;
}

/* Compile with: gcc program.c -lm */

```

8. Square of a number

```

#include <stdio.h>

int main() {
    float n;
    scanf("%f", &n);
    printf("Square = %.2f", n * n);
    return 0;
}

```

9. Area of a circle

```

#include <stdio.h>

int main() {
    float r, area;
    const float PI = 3.14159;

    scanf("%f", &r);
    area = PI * r * r;

    printf("Area = %.2f", area);
    return 0;
}

```

10. Area of a triangle

```

#include <stdio.h>

int main() {
    float base, height, area;
    scanf("%f %f", &base, &height);

    area = 0.5 * base * height;
    printf("Area = %.2f", area);

    return 0;
}

```

11. Convert minutes to hours and minutes

```

#include <stdio.h>

int main() {
    int minutes, hours, remaining;
    scanf("%d", &minutes);

    hours = minutes / 60;
    remaining = minutes % 60;

    printf("%d hours %d minutes", hours, remaining);
    return 0;
}

```

B. Conditional Statements

12. Largest of three numbers

```
#include <stdio.h>

int main() {
    int a, b, c, largest;
    scanf("%d %d %d", &a, &b, &c);

    if (a >= b && a >= c)
        largest = a;
    else if (b >= a && b >= c)
        largest = b;
    else
        largest = c;

    printf("Largest = %d", largest);
    return 0;
}
```

13. Smallest of three numbers

```
#include <stdio.h>

int main() {
    int a, b, c, smallest;
    scanf("%d %d %d", &a, &b, &c);

    if (a <= b && a <= c)
        smallest = a;
    else if (b <= a && b <= c)
        smallest = b;
    else
        smallest = c;

    printf("Smallest = %d", smallest);
    return 0;
}
```

14. Even or odd

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    if (n % 2 == 0)
        printf("Even");
    else
        printf("Odd");

    return 0;
}
```

15. Leap year

```
#include <stdio.h>

int main() {
    int year;
    scanf("%d", &year);

    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
        printf("Leap year");
    else
        printf("Not a leap year");

    return 0;
}
```

16. Letter grade to number grade

```
#include <stdio.h>

int main() {
    char grade;
    scanf(" %c", &grade);
}
```

```

switch (grade) {
    case 'A': case 'a':
        printf("Number Grade = 4.00");
        break;
    case 'B': case 'b':
        printf("Number Grade = 3.00");
        break;
    case 'C': case 'c':
        printf("Number Grade = 2.00");
        break;
    case 'D': case 'd':
        printf("Number Grade = 1.00");
        break;
    case 'F': case 'f':
        printf("Number Grade = 0.00");
        break;
    default:
        printf("Invalid grade");
}

return 0;
}

```

17. Simple calculator using switch-case

```

#include <stdio.h>

int main() {
    float a, b;
    char op;

    scanf("%f %c %f", &a, &op, &b);

    switch (op) {
        case '+':
            printf("%.2f", a + b);
            break;
        case '-':
            printf("%.2f", a - b);
            break;
        case '*':
            printf("%.2f", a * b);
            break;
        case '/':
            if (b != 0)
                printf("%.2f", a / b);
            else
                printf("Cannot divide by zero");
            break;
        default:
            printf("Invalid operator");
    }

    return 0;
}

```

18. Second largest number

```

#include <stdio.h>

int main() {
    int a, b, c, largest, second;
    scanf("%d %d %d", &a, &b, &c);

    if (a >= b && a >= c) {
        largest = a;
        second = (b >= c) ? b : c;
    }
    else if (b >= a && b >= c) {
        largest = b;
        second = (a >= c) ? a : c;
    }
    else {
        largest = c;
        second = (a >= b) ? a : b;
    }

    printf("Second largest = %d", second);
    return 0;
}

```

```
}
```

19. Check triangle type

```
#include <stdio.h>

int main() {
    int a, b, c;
    scanf("%d %d %d", &a, &b, &c);

    if (a + b <= c || a + c <= b || b + c <= a)
        printf("Invalid triangle");
    else if (a == b && b == c)
        printf("Equilateral");
    else if (a == b || b == c || a == c)
        printf("Isosceles");
    else
        printf("Scalene");

    return 0;
}
```

20. Negative, positive or zero

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    if (n > 0)
        printf("Positive");
    else if (n < 0)
        printf("Negative");
    else
        printf("Zero");

    return 0;
}
```

21. Vowel or consonant using switch-case

```
#include <stdio.h>

int main() {
    char ch;
    scanf(" %c", &ch);

    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            printf("Vowel");
            break;
        default:
            printf("Consonant");
    }

    return 0;
}
```

C. Loops and Number Programs

22. Multiplication table using while loop

```
#include <stdio.h>

int main() {
    int n, i = 1;
    scanf("%d", &n);

    while (i <= 10) {
        printf("%d x %d = %d\n", n, i, n * i);
        i++;
    }

    return 0;
}
```

23. Fibonacci series using for loop

```
#include <stdio.h>

int main() {
    int n, i;
    int a = 0, b = 1, c;

    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        printf("%d ", a);
        c = a + b;
        a = b;
        b = c;
    }

    return 0;
}
```

24. Sum and average of first 100 natural numbers

```
#include <stdio.h>

int main() {
    int i = 1, sum = 0;
    float average;

    do {
        sum += i;
        i++;
    } while (i <= 100);

    average = sum / 100.0;

    printf("Sum = %d\n", sum);
    printf("Average = %.2f", average);

    return 0;
}
```

25. Check whether a number is prime

```
#include <stdio.h>

int main() {
    int n, i, prime = 1;
    scanf("%d", &n);

    if (n < 2)
        prime = 0;

    for (i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            prime = 0;
            break;
        }
    }

    if (prime)
        printf("Prime");
}
```

```

        else
            printf("Not Prime");

    return 0;
}

```

26. Factorial using while loop

```

#include <stdio.h>

int main() {
    int n, i = 1;
    long long fact = 1;

    scanf("%d", &n);

    while (i <= n) {
        fact *= i;
        i++;
    }

    printf("Factorial = %lld", fact);
    return 0;
}

```

27. Print a series (1 to n)

```

#include <stdio.h>

int main() {
    int n, i;
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
        printf("%d ", i);

    return 0;
}

```

28. Triangle, reversed triangle and pyramid patterns

```

#include <stdio.h>

int main() {
    int n, i, j;
    scanf("%d", &n);

    /* Triangle */
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }

    printf("\n");

    /* Reversed triangle */
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }

    printf("\n");

    /* Pyramid */
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++)
            printf(" ");

        for (j = 1; j <= 2 * i - 1; j++)
            printf("*");

        printf("\n");
    }

    return 0;
}

```

29. Convert decimal number to another base

```
#include <stdio.h>

int main() {
    int n, base, result[100], i = 0;
    scanf("%d %d", &n, &base);

    if (base < 2 || base > 16) {
        printf("Base must be between 2 and 16");
        return 0;
    }

    if (n == 0) {
        printf("0");
        return 0;
    }

    while (n > 0) {
        result[i++] = n % base;
        n /= base;
    }

    while (i > 0) {
        i--;

        if (result[i] < 10)
            printf("%d", result[i]);
        else
            printf("%c", 'A' + result[i] - 10);
    }

    return 0;
}
```

30. Display cube of numbers up to an integer

```
#include <stdio.h>

int main() {
    int n, i;
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
        printf("Cube of %d = %d\n", i, i * i * i);

    return 0;
}
```

31. Armstrong numbers in a range

```
#include <stdio.h>

int main() {
    int start, end, n, temp, digit, sum;
    scanf("%d %d", &start, &end);

    for (n = start; n <= end; n++) {
        temp = n;
        sum = 0;

        while (temp > 0) {
            digit = temp % 10;
            sum += digit * digit * digit;
            temp /= 10;
        }

        if (sum == n)
            printf("%d ", n);
    }

    return 0;
}

/* This version uses the classic 3-digit Armstrong definition. */
```

Note: This implementation uses the classic 3-digit Armstrong definition.

32. Sum of digits of a number

```
#include <stdio.h>

int main() {
    int n, sum = 0, digit;
    scanf("%d", &n);

    if (n < 0)
        n = -n;

    while (n > 0) {
        digit = n % 10;
        sum += digit;
        n /= 10;
    }

    printf("Sum = %d", sum);
    return 0;
}
```

D. Arrays

33. Sum of elements in an array

```
#include <stdio.h>

int main() {
    int a[100], n, i, sum = 0;
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        sum += a[i];
    }

    printf("Sum = %d", sum);
    return 0;
}
```

34. Store and print array elements

```
#include <stdio.h>

int main() {
    int a[100], n, i;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

35. Display array in reverse order

```
#include <stdio.h>

int main() {
    int a[100], n, i;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = n - 1; i >= 0; i--)
        printf("%d ", a[i]);

    return 0;
}
```

36. Maximum and minimum elements in an array

```
#include <stdio.h>

int main() {
    int a[100], n, i, max, min;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    max = min = a[0];

    for (i = 1; i < n; i++) {
        if (a[i] > max)
            max = a[i];

        if (a[i] < min)
            min = a[i];
    }

    printf("Maximum = %d\n", max);
    printf("Minimum = %d", min);

    return 0;
}
```

37. Count negative elements in an array

```
#include <stdio.h>

int main() {
    int a[100], n, i, count = 0;
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);

        if (a[i] < 0)
            count++;
    }

    printf("Negative elements = %d", count);
    return 0;
}
```

38. Find the middle array index

```
#include <stdio.h>

int main() {
    int a[100], n, i;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    if (n % 2 == 1)
        printf("Middle index = %d", n / 2);
    else
        printf("Middle indices = %d and %d", n / 2 - 1, n / 2);

    return 0;
}
```

39. Copy one array into another

```
#include <stdio.h>

int main() {
    int a[100], b[100], n, i;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++)
        b[i] = a[i];

    for (i = 0; i < n; i++)
        printf("%d ", b[i]);

    return 0;
}
```

40. Count duplicate elements in an array

```
#include <stdio.h>

int main() {
    int a[100], n, i, j, count = 0;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                count++;
                break;
            }
        }
    }

    printf("Duplicate elements = %d", count);
}
```

```

    return 0;
}

```

41. Frequency of each array element

```

#include <stdio.h>

int main() {
    int a[100], visited[100] = {0};
    int n, i, j, count;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++) {
        if (visited[i])
            continue;

        count = 1;

        for (j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                count++;
                visited[j] = 1;
            }
        }

        printf("%d -> %d\n", a[i], count);
    }

    return 0;
}

```

42. Average of array elements

```

#include <stdio.h>

int main() {
    int a[100], n, i, sum = 0;
    float average;

    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        sum += a[i];
    }

    average = (float)sum / n;
    printf("Average = %.2f", average);

    return 0;
}

```

43. Sort array in ascending order

```

#include <stdio.h>

int main() {
    int a[100], n, i, j, temp;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (a[i] > a[j]) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }

    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
}

```

```

    return 0;
}

```

44. Sort array in descending order

```

#include <stdio.h>

int main() {
    int a[100], n, i, j, temp;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (a[i] < a[j]) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }

    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}

```

45. Search for an element in an array

```

#include <stdio.h>

int main() {
    int a[100], n, i, key, found = 0;
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &key);

    for (i = 0; i < n; i++) {
        if (a[i] == key) {
            printf("Element found at index %d", i);
            found = 1;
            break;
        }
    }

    if (!found)
        printf("Element not found");

    return 0;
}

```

E. Strings

46. Input and display a string

```
#include <stdio.h>

int main() {
    char str[100];
    fgets(str, sizeof(str), stdin);
    printf("%s", str);
    return 0;
}
```

47. Concatenate two strings without strcat()

```
#include <stdio.h>

int main() {
    char a[200], b[100];
    int i, j;

    fgets(a, sizeof(a), stdin);
    fgets(b, sizeof(b), stdin);

    for (i = 0; a[i] != '\0'; i++);

    if (i > 0 && a[i - 1] == '\n')
        i--;

    for (j = 0; b[j] != '\0' && i + j < 199; j++) {
        if (b[j] != '\n')
            a[i + j] = b[j];
    }

    a[i + j] = '\0';
    printf("%s", a);

    return 0;
}
```

48. Check whether two strings are equal

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[100], b[100];

    fgets(a, sizeof(a), stdin);
    fgets(b, sizeof(b), stdin);

    if (strcmp(a, b) == 0)
        printf("Equal");
    else
        printf("Not Equal");

    return 0;
}
```

49. Reverse a string

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i;

    fgets(str, sizeof(str), stdin);

    for (i = (int)strlen(str) - 1; i >= 0; i--) {
        if (str[i] != '\n')
            printf("%c", str[i]);
    }

    return 0;
}
```

50. Count vowels in a string

```
#include <stdio.h>

int main() {
    char str[200];
    int i, count = 0;

    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++) {
        switch (str[i]) {
            case 'a': case 'e': case 'i': case 'o': case 'u':
            case 'A': case 'E': case 'I': case 'O': case 'U':
                count++;
        }
    }

    printf("Vowels = %d", count);
    return 0;
}
```

51. Count words in a string

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[200];
    int i, words = 0, inWord = 0;

    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++) {
        if (!isspace((unsigned char)str[i]) && !inWord) {
            words++;
            inWord = 1;
        }
        else if (isspace((unsigned char)str[i])) {
            inWord = 0;
        }
    }

    printf("Words = %d", words);
    return 0;
}
```

52. String length without strlen()

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0' && str[i] != '\n')
        i++;

    printf("Length = %d", i);
    return 0;
}
```

53. Convert string to uppercase

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    int i;

    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
        str[i] = (char)toupper((unsigned char)str[i]);

    printf("%s", str);
}
```

```

    return 0;
}

```

54. Convert string to lowercase

```

#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    int i;

    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
        str[i] = (char)tolower((unsigned char)str[i]);

    printf("%s", str);
    return 0;
}

```

55. Extract a substring

```

#include <stdio.h>

int main() {
    char str[200], sub[100];
    int start, length, i;

    fgets(str, sizeof(str), stdin);
    scanf("%d %d", &start, &length);

    for (i = 0; i < length && str[start + i] != '\0'; i++)
        sub[i] = str[start + i];

    sub[i] = '\0';
    printf("%s", sub);

    return 0;
}

```

56. Remove all spaces from a string

```

#include <stdio.h>

int main() {
    char str[200];
    int i, j = 0;

    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] != ' ')
            str[j++] = str[i];
    }

    str[j] = '\0';
    printf("%s", str);

    return 0;
}

```

57. Replace a character in a string

```

#include <stdio.h>

int main() {
    char str[200], oldChar, newChar;
    int i;

    fgets(str, sizeof(str), stdin);
    scanf(" %c %c", &oldChar, &newChar);

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] == oldChar)
            str[i] = newChar;
    }

    printf("%s", str);
    return 0;
}

```

```
}
```

58. Sort strings alphabetically

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[10][100], temp[100];
    int n, i, j;

    scanf("%d", &n);
    getchar();

    for (i = 0; i < n; i++)
        fgets(str[i], sizeof(str[i]), stdin);

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (strcmp(str[i], str[j]) > 0) {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    for (i = 0; i < n; i++)
        printf("%s", str[i]);

    return 0;
}
```

59. Check whether a string is a palindrome

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int left = 0, right, palindrome = 1;

    fgets(str, sizeof(str), stdin);
    right = (int)strlen(str) - 1;

    if (right >= 0 && str[right] == '\n')
        right--;

    while (left < right) {
        if (str[left] != str[right]) {
            palindrome = 0;
            break;
        }

        left++;
        right--;
    }

    if (palindrome)
        printf("Palindrome");
    else
        printf("Not Palindrome");

    return 0;
}
```

60. Remove all occurrences of a character

```
#include <stdio.h>

int main() {
    char str[200], ch;
    int i, j = 0;

    fgets(str, sizeof(str), stdin);
    scanf(" %c", &ch);

    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] != ch)
            str[j++] = str[i];
    }
}
```

```
    }  
    str[j] = '\\0';  
    printf("%s", str);  
    return 0;  
}
```

F. Functions and Recursion

61. Cube using a function

```
#include <stdio.h>

int cube(int n) {
    return n * n * n;
}

int main() {
    int n;
    scanf("%d", &n);

    printf("Cube = %d", cube(n));
    return 0;
}
```

62. Maximum and minimum using functions

```
#include <stdio.h>

int maximum(int a, int b) {
    return a > b ? a : b;
}

int minimum(int a, int b) {
    return a < b ? a : b;
}

int main() {
    int a, b;
    scanf("%d %d", &a, &b);

    printf("Maximum = %d\n", maximum(a, b));
    printf("Minimum = %d", minimum(a, b));

    return 0;
}
```

63. Print numbers between intervals using a function

```
#include <stdio.h>

void printNumbers(int start, int end) {
    int i;

    for (i = start; i <= end; i++)
        printf("%d ", i);
}

int main() {
    int start, end;
    scanf("%d %d", &start, &end);

    printNumbers(start, end);
    return 0;
}
```

64. GCD using a function

```
#include <stdio.h>

int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }

    return a;
}

int main() {
    int a, b;
    scanf("%d %d", &a, &b);

    printf("GCD = %d", gcd(a, b));
    return 0;
}
```

```
}
```

65. LCM using a function

```
#include <stdio.h>

int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }

    return a;
}

int lcm(int a, int b) {
    return (a / gcd(a, b)) * b;
}

int main() {
    int a, b;
    scanf("%d %d", &a, &b);

    printf("LCM = %d", lcm(a, b));
    return 0;
}
```

66. Area of a circle using a function

```
#include <stdio.h>

float area(float r) {
    return 3.14159f * r * r;
}

int main() {
    float r;
    scanf("%f", &r);

    printf("Area = %.2f", area(r));
    return 0;
}
```

67. Celsius to Fahrenheit using a function

```
#include <stdio.h>

float fahrenheit(float c) {
    return (c * 9 / 5) + 32;
}

int main() {
    float c;
    scanf("%f", &c);

    printf("Fahrenheit = %.2f", fahrenheit(c));
    return 0;
}
```

68. Factorial using a function

```
#include <stdio.h>

long long factorial(int n) {
    long long fact = 1;
    int i;

    for (i = 1; i <= n; i++)
        fact *= i;

    return fact;
}

int main() {
    int n;
    scanf("%d", &n);

    printf("Factorial = %lld", factorial(n));
    return 0;
}
```

```
}
```

69. Power using recursion

```
#include <stdio.h>

long long power(int base, int exponent) {
    if (exponent == 0)
        return 1;

    return base * power(base, exponent - 1);
}

int main() {
    int base, exponent;
    scanf("%d %d", &base, &exponent);

    printf("Result = %lld", power(base, exponent));
    return 0;
}
```

70. Sum of natural numbers 1 to n using recursion

```
#include <stdio.h>

int sum(int n) {
    if (n == 0)
        return 0;

    return n + sum(n - 1);
}

int main() {
    int n;
    scanf("%d", &n);

    printf("Sum = %d", sum(n));
    return 0;
}
```

71. Reverse a number using recursion

```
#include <stdio.h>

int reverse(int n, int rev) {
    if (n == 0)
        return rev;

    return reverse(n / 10, rev * 10 + n % 10);
}

int main() {
    int n;
    scanf("%d", &n);

    printf("Reverse = %d", reverse(n, 0));
    return 0;
}
```

72. nth Fibonacci term using recursion

```
#include <stdio.h>

int fibonacci(int n) {
    if (n == 0)
        return 0;

    if (n == 1)
        return 1;

    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main() {
    int n;
    scanf("%d", &n);

    printf("Fibonacci term = %d", fibonacci(n));
    return 0;
}
```

G. Pointers

73. Swap two numbers using pointers

```
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int a, b;
    scanf("%d %d", &a, &b);

    swap(&a, &b);

    printf("After swapping: %d %d", a, b);
    return 0;
}
```

74. Reverse a string using pointers

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], *start, *end, temp;

    fgets(str, sizeof(str), stdin);

    start = str;
    end = str + strlen(str) - 1;

    if (*end == '\n')
        end--;

    while (start < end) {
        temp = *start;
        *start = *end;
        *end = temp;

        start++;
        end--;
    }

    printf("%s", str);
    return 0;
}
```

75. Largest array element using pointers

```
#include <stdio.h>

int main() {
    int a[100], n, i, largest;
    int *p;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    p = a;
    largest = *p;

    for (i = 1; i < n; i++) {
        if (*(p + i) > largest)
            largest = *(p + i);
    }

    printf("Largest = %d", largest);
    return 0;
}
```

76. Smallest array element using pointers

```
#include <stdio.h>
```

```

int main() {
    int a[100], n, i, smallest;
    int *p;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    p = a;
    smallest = *p;

    for (i = 1; i < n; i++) {
        if (*(p + i) < smallest)
            smallest = *(p + i);
    }

    printf("Smallest = %d", smallest);
    return 0;
}

```

77. Sort an array using pointers

```

#include <stdio.h>

int main() {
    int a[100], n, i, j, temp;
    int *p = a;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", p + i);

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (*(p + i) > *(p + j)) {
                temp = *(p + i);
                *(p + i) = *(p + j);
                *(p + j) = temp;
            }
        }
    }

    for (i = 0; i < n; i++)
        printf("%d ", *(p + i));

    return 0;
}

```

H. File Handling

78. Read and display contents of a file

```
#include <stdio.h>

int main() {
    FILE *file;
    char ch;

    file = fopen("input.txt", "r");

    if (file == NULL) {
        printf("File not found");
        return 1;
    }

    while ((ch = fgetc(file)) != EOF)
        putchar(ch);

    fclose(file);
    return 0;
}
```

79. Write data to a file

```
#include <stdio.h>

int main() {
    FILE *file;
    char text[200];

    file = fopen("output.txt", "w");

    if (file == NULL) {
        printf("Unable to create file");
        return 1;
    }

    fgets(text, sizeof(text), stdin);
    fprintf(file, "%s", text);

    fclose(file);

    printf("Data written successfully");
    return 0;
}
```

80. Copy contents of one file to another

```
#include <stdio.h>

int main() {
    FILE *source, *destination;
    char ch;

    source = fopen("source.txt", "r");
    destination = fopen("copy.txt", "w");

    if (source == NULL || destination == NULL) {
        printf("Unable to open file");
        return 1;
    }

    while ((ch = fgetc(source)) != EOF)
        fputc(ch, destination);

    fclose(source);
    fclose(destination);

    printf("File copied successfully");
    return 0;
}
```

81. Count words in a file

```
#include <stdio.h>
#include <ctype.h>
```

```

int main() {
    FILE *file;
    char ch;
    int words = 0, inWord = 0;

    file = fopen("input.txt", "r");

    if (file == NULL) {
        printf("File not found");
        return 1;
    }

    while ((ch = fgetc(file)) != EOF) {
        if (!isspace((unsigned char)ch) && !inWord) {
            words++;
            inWord = 1;
        }
        else if (isspace((unsigned char)ch)) {
            inWord = 0;
        }
    }

    fclose(file);

    printf("Number of words = %d", words);
    return 0;
}

```

82. Encrypt and decrypt a file

```

#include <stdio.h>

void encryptFile() {
    FILE *source, *destination;
    char ch;

    source = fopen("input.txt", "r");
    destination = fopen("encrypted.txt", "w");

    if (source == NULL || destination == NULL) {
        printf("Unable to open file");
        return;
    }

    while ((ch = fgetc(source)) != EOF) {
        ch = ch + 3;
        fputc(ch, destination);
    }

    fclose(source);
    fclose(destination);

    printf("File encrypted successfully.\n");
}

void decryptFile() {
    FILE *source, *destination;
    char ch;

    source = fopen("encrypted.txt", "r");
    destination = fopen("decrypted.txt", "w");

    if (source == NULL || destination == NULL) {
        printf("Unable to open file");
        return;
    }

    while ((ch = fgetc(source)) != EOF) {
        ch = ch - 3;
        fputc(ch, destination);
    }

    fclose(source);
    fclose(destination);

    printf("File decrypted successfully.\n");
}

int main() {

```

```
int choice;

printf("1. Encrypt\n");
printf("2. Decrypt\n");
scanf("%d", &choice);

if (choice == 1)
    encryptFile();
else if (choice == 2)
    decryptFile();
else
    printf("Invalid choice");

return 0;
}
```

Note: This is a simple Caesar-style file transformation for lab demonstration, not modern secure encryption.